Practical 06: Communication Between Activity and Fragment

Source: Android Programming Practical(1).pdf | Source pages: 46–53

Aim

To develop and test an Android application demonstrating communication between activity and fragment.

Steps to Perform the Practical

1. Create a new Android Studio project for Activity–Fragment communication.

2. Create the fragment class and its layout using the names in the source practical.

3. Add the fragment to the activity and implement the communication interface or callback method shown in the supplied code.

4. Pass or return the required data between the Activity and Fragment as demonstrated.

5. Run the app and perform the input/action that triggers communication.

6. Verify that the received value is displayed in the correct screen or view.

Result

The application is run and the expected behavior is verified according to the practical instructions and code.

Note: Use the package name and resource names from the source code consistently. If Android Studio reports a missing resource or class, check spelling and ensure all files are in the correct folders.

6. Develop an Android application demonstrating communication between Activity and Fragment.

Project Name: ActivityFragmentCommunication

Creating Fragment
java package → New→ Fragment → Fragment (Blank)
Fragment Name: DemoFragment
Layout Name: fragment_demo

Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<FrameLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

   android:id="@+id/fragment_container"

   android:layout_width="match_parent"

   android:layout_height="match_parent"/>
```

Open From res/layout:
Filename: fragment_demo.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

   android:layout_width="match_parent"

   android:layout_height="match_parent"

   android:gravity="center"

   android:orientation="vertical"

   android:padding="20dp">



   <TextView
```

```
       android:id="@+id/txtMessage"

       android:layout_width="wrap_content" android:layout_height="wrap_content"
android:textSize="22sp"

       android:text="Message from Activity"/>



   <Button

       android:id="@+id/btnSend"

       android:layout_width="wrap_content"

       android:layout_height="wrap_content"

       android:text="Send Message to Activity"

       android:layout_marginTop="20dp"/>

</LinearLayout>
```

Modify DemoFragment.java.

Filename: DemoFragment.java

```
package com.example.activityfragmentcommunication;


import android.content.Context;

import android.os.Bundle;

import android.view.LayoutInflater;

import android.view.View;

import android.view.ViewGroup; import android.widget.Button;

import android.widget.TextView;

import androidx.annotation.NonNull;

import androidx.annotation.Nullable;
```

```
import androidx.fragment.app.Fragment;

public class DemoFragment extends Fragment

{

   TextView txtMessage;

   Button btnSend;

   String activityMessage;



   FragmentCommunication communication;

public interface FragmentCommunication {

   void onMessageFromFragment(String message);

}



// Receive Activity object

@Override

public void onAttach(@NonNull Context context) {

   super.onAttach(context);

   communication = (FragmentCommunication) context;

}







// Method to receive data from Activity

public static DemoFragment newInstance(String message) {

   DemoFragment fragment = new DemoFragment();

   Bundle bundle = new Bundle();

   bundle.putString("msg", message);
```

```
    fragment.setArguments(bundle);

    return fragment;

}




@Override

public View onCreateView(LayoutInflater inflater,ViewGroup container,Bundle
savedInstanceState) {

    View view = inflater.inflate(R.layout.fragment_demo,container, false);

    txtMessage = view.findViewById(R.id.txtMessage);

    btnSend = view.findViewById(R.id.btnSend);

// Receiving data from Activity

    if(getArguments() != null){

        activityMessage = getArguments().getString("msg");

        txtMessage.setText(activityMessage);

    }



// Sending data to Activity

    btnSend.setOnClickListener(v -> {

        communication.onMessageFromFragment(

                "Hello Activity! Message from Fragment");

    }

    );



        return view;

}
```

```
}
```

Filename: MainActivity.java

```
package com.example.activityfragmentcommunication;


import android.os.Bundle;

import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;


public class MainActivity extends AppCompatActivity

       implements DemoFragment.FragmentCommunication {


   @Override

   protected void onCreate(Bundle savedInstanceState) {

       super.onCreate(savedInstanceState);

       setContentView(R.layout.activity_main);


       // Sending data from Activity to Fragment

       DemoFragment fragment = DemoFragment.newInstance(

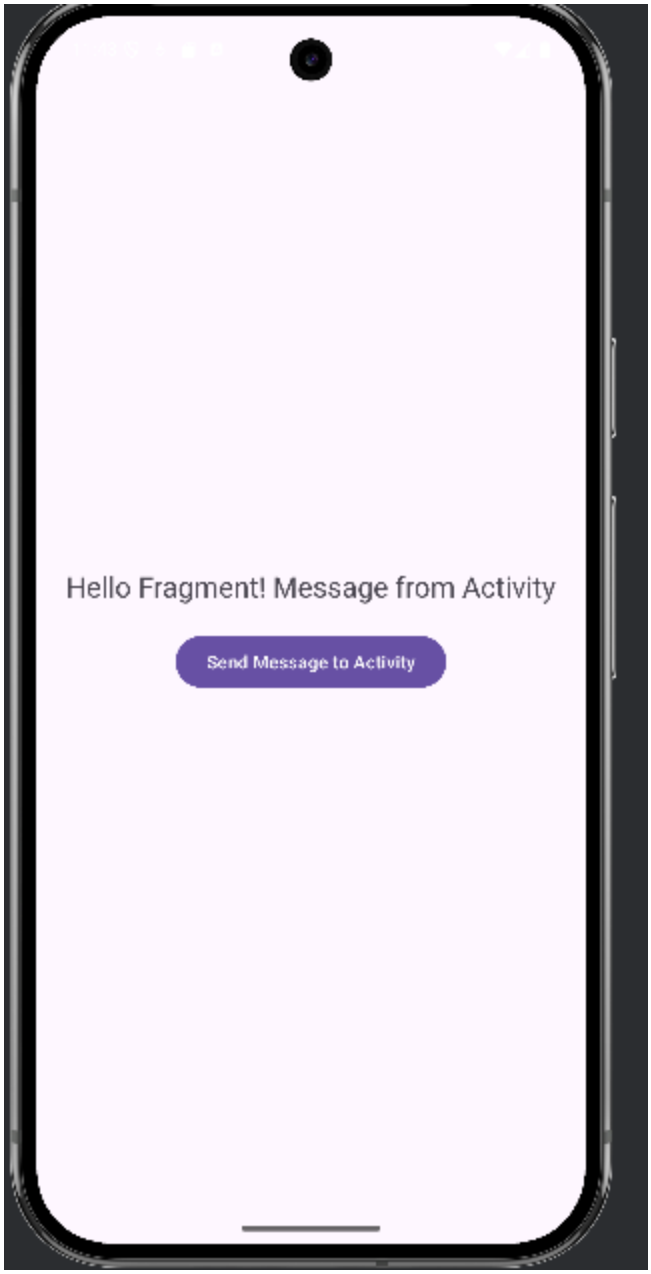
               "Hello Fragment! Message from Activity"
```

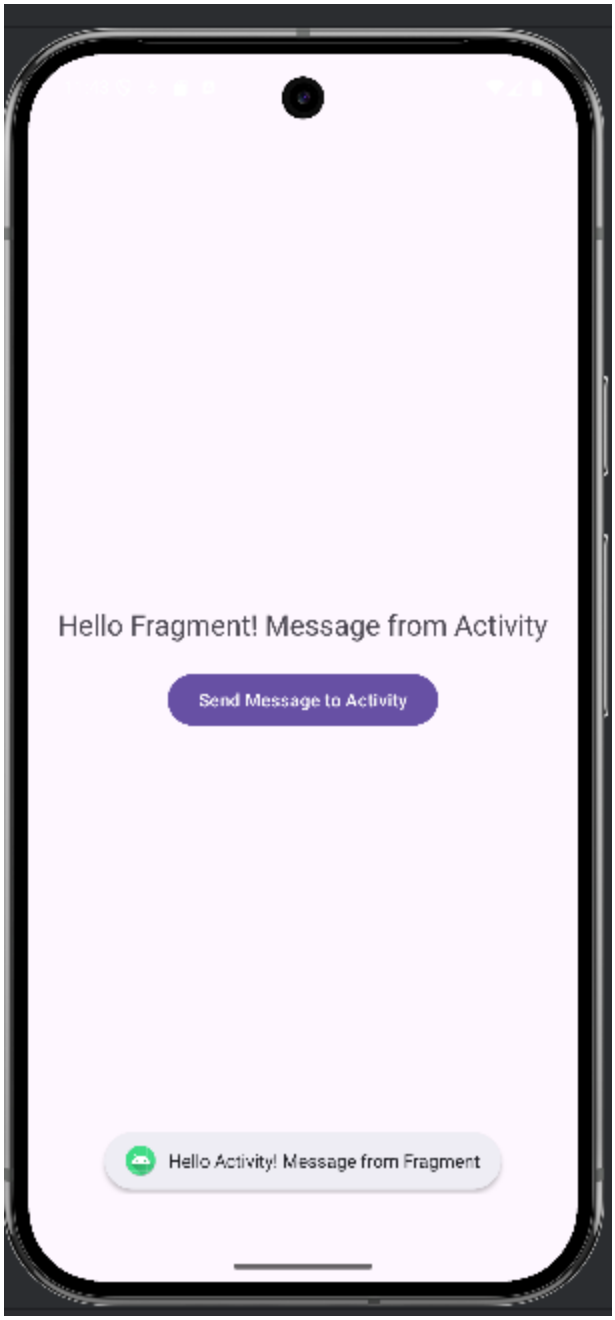
```
        );


        getSupportFragmentManager()
                .beginTransaction()
                .replace(R.id.fragment_container, fragment)
                .commit();
    }


    // Receiving data from Fragment
    @Override
    public void onMessageFromFragment(String message) {
        Toast.makeText(this, message, Toast.LENGTH_LONG).show();
    }
}
```

OUTPUT:


Hello Fragment! Message from Activity
Send Message to Activity


Hello Fragment! Message from Activity
Send Message to Activity
Hello Activity! Message from Fragment